

Lia Schmid

B.Sc. COGNITIVE SCIENCE · M.Sc. SCIENTIFIC COMPUTING

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Summary

Driven by a vision of using technology to enhance societal well-being, I aspire to bridge health and computer science in meaningful ways. With an interdisciplinary academic journey, I seek to create innovative solutions that connect diverse fields, driving research and innovation forward while keeping social impact at the heart of my work. I hope to work together in enthusiastic teams to learn things from each other.

Education

(European) Sport Nutritionist

ACCADEMIA FITNESS ITALIANA

Milan, Italy

12/2025 – Today

- Certified Sports Nutritionist: Trained in human physiology, sports psychology, and the application of professional nutritional plans.
- Client Management Specialist: Developed practical skills for designing dietary strategies and managing professional athlete-client relations.

Master of Science in Scientific Computing

RUPRECHT-KARLS-UNIVERSITÄT

Heidelberg, Germany

04/2021 – 11/2024

- Thesis: "Automatic Segmentation of Choroid Plexus in Neurodegenerative Diseases: Leveraging the T1xFLAIR Product as a Novel Ground Truth Modality" (Collaboration: University of Milan, Policlinico of Milan and University of Heidelberg).
- Study abroad (4EU+ Erasmus programme): Year abroad at Università degli Studi di Milano, Milan, Italy (09/2022 – 09/2023)
- Final Degree: 1,1

Bachelor of Science in Cognitive Science

EBERHARD KARLS UNIVERSITÄT

Tübingen, Germany

10/2017 – 04/2021

- Study abroad (Erasmus programme): Semester abroad at Universidade Nova de Lisboa, Lisbon, Portugal (09/2019 – 01/2020)
- Final Degree: 1,47

Work Experience

Perceptual computing and Human Sensing Lab (PHuSe Lab), University of Milan

RESEARCH FELLOW (POSTGRADUATE RESEARCHER) IN BRAIN-COMPUTER INTERFACES (EEG)

Milan, Italy

04/2025 – today

- Conduct research on EEG-based Brain-Computer Interfaces (BCIs).
- Develop and evaluate machine learning models for decoding brain signals and user intent.

Catholic University of the Sacred Heart

TUTOR FOR COMPUTER PROGRAMMING AND DATABASE SYSTEMS

Milan, Italy

02/2026 – today

- Exercitation for the M.Sc. Data Science and AI for Business
- Python Programming and DBMS (MySQL, DBeaver)

Engineering Mathematics and Computing Lab (EMCL), University of Heidelberg

RESEARCH ASSISTANT

Heidelberg, Germany

05/2023 – 02/2024

- Co-organizing a project to analyze morphological structures such as those of fish.
- Setting up Singularity container on the University server.

Eberhard Karls University

TUTOR IN MATHEMATICS FOR COMPUTER SCIENCE

Tübingen, Germany

10/2020 – 12/2020

- Preparation course for freshmen.
- Assisting new students in achieving a consistent mathematical foundation while building social connections.

Vision and Cognition Lab, Werner Reichardt Centre for Integrative Neuroscience

RESEARCH INTERNSHIP/ RESEARCH ASSISTANT

Tübingen, Germany

03/2020 – 12/2020

- Safety briefing on an fMRI device.
- Comparing hierarchical processing of the visual cortex to layers of convolutional neural networks (python, MATLAB).

Eberhard Karls University

TUTOR IN MATHEMATICS FOR COMPUTER SCIENCE

Tübingen, Germany

10/2020 – 12/2020

- Covering topics of calculus, linear algebra, optimization, and discrete mathematics.

- Providing peer-to-peer guidance on course selection, schedule planning, and exam requirements.
- Assisting students with organizational questions, including housing and funding options.

Publications

- Schmid, L., Burger, J., D'Amelio, Lanzarotti, R.: "Investigating Foundation Models, Disentanglement and Latent Alignment for Subject-Independent EEG Learning" (Submitted: ICMI 2026).
- Schmid, L., Facchi, G., Agnelli, F., Bocca, G., Sacchi, L., Lanzarotti, R.: "Choroid Plexus Segmentation in MRI Using the Novel T1xFLAIR Modality and PSU-Mamba: Projective Scan U-Mamba Approach" (PRL Elsevier 2025).
- Sacchi, L., Arcaro M., Bocca, G., Schmid, L., Carandini T., Ghezzi L., Pintus M., Pietroboni A. M., Fenoglio C., Serpente M., Conte G., Triulzi F., Lanzarotti R., Dolci C., Galimberti, D., and Arighi A. "Klotho levels in the cerebrospinal fluid are associated with choroid plexus enlargement in neurodegeneration: a preliminary study" (Frontiers in Aging Neuroscience)
- Agnelli, F., Ghezzi, O., Blandano, G., Burger, J., Facchi, G., and Schmid, L. Enhancing 3D Face Analysis Using Graph Convolutional Networks with Kernel-Attentive Filters. Manuscript submitted to The 40th ACM/SIGAPP Symposium on Applied Computing (SAC 2025).
- Agnelli, F., Blandano, G., Burger, J., D'Amelio, A., Facchi, G., Ghezzi, O., ... & Schmid, L. (2025, September). EEG-Based Mental Stress Detection: A Comparative and Explainable Study Across Tasks and Subjects (ICIAP 2025).

Projects

Development of a BCI solution based on the acquisition and analysis of EEG signals

Milan, Italy

BRAINCAPNET

01/2024 - 11/2024

- Developed the DVAE model, integrating CBRAMOD as a feature extractor alongside a multi-branch U-Net-like bottleneck cascade to process spatial and temporal dimensions.
- Conducting ongoing data collection to compile a new P300 dataset involving healthy adult subjects.
- Engineered an online streaming analyzer for real-time EEG buffering, filtering, and predictive classification.
- Developed and field-tested three neurorehabilitative games via PsychoPy in collaboration with Dotdotdot and Fondazione TOG.

Automatic Segmentation of Choroid Plexus in Neurodegenerative Diseases: Leveraging the T1xFLAIR Product as a Novel Ground Truth Modality

Milan, Italy

MASTER'S THESIS

01/2024 - 11/2024

- Developed a novel approach for choroid plexus segmentation using 3D MRI data from patients with neurodegenerative pathologies.
- Created and validated the T1xFLAIR modality as a new ground truth for segmentation when contrast-enhanced T1w imaging is unavailable.
- Designed and trained machine learning models for automatic segmentation, optimizing for clinical applicability.
- Skills acquired: MRI data preprocessing, manual segmentation, neuroimaging techniques, machine learning model development, Python (PyTorch, MONAI), and interdisciplinary collaboration.

Investigating Shape and Texture Bias in Neural Networks and the Human Visual Cortex

Tübingen, Germany

INTERNSHIP: VISION AND COGNITION LAB, WERNER REICHARDT CENTRE FOR INTEGRATIVE NEUROSCIENCE

03/2020 - 12/2020

- Explored the relationship between convolutional neural networks (CNNs) and the human visual cortex through an fMRI experiment.
- Conducted an fMRI experiment where participants viewed images generated using neural style transfer to isolate texture and shape cues.
- Analyzed correlations between activations of visual cortex regions of interest (ROIs) and layers of trained CNNs to explore similarities in hierarchical processing.
- Skills acquired: fMRI data analysis, neural network interpretation, Python (e.g., PyTorch, NumPy, and statistical packages), Matlab and interdisciplinary research in computational neuroscience.

Selective Temporal Attention using EEG signals

Tübingen, Germany

UNIVERSITY TEAM PROJECT

05/2019 - 09/2019

- Conducted EEG data collection using a cap-based system to investigate the modulation of early auditory processing by spatial and temporal attention.
- Applied the Hillyard paradigm (selective attention task using event-related potentials) to study multidimensional selection in auditory and visual modalities.
- Findings: Temporal and spatial attention enhance early ERP components (N1) and interact within the same auditory processing stages.

Technical Skills

- **Programming Languages:** Python, R, Matlab, bash, Java.
- **Databases:** MySQL
- **Developer Tools:** git, VS Code, Singularity.

Extracurricular Activity

Wizz Air Marathon Milano 2026

MEDAL ENGRAVING AND PACERS BAGGAGE

- Medal Engraving, pacers baggage, dog run support

Milan, Italy

04/2026

Milano Cortina 26

EVENT SERVICE VOLUNTEER OLYMPIC GAMES MILANO CORTINA 26

- Event Service at Olympic Village, Milan

Milan, Italy

01/2026 - 02/2026

University of Tübingen

MEMBER OF THE STUDY COMMISSION OF COGNITIVE SCIENCE

- Development and evaluation of study plan
- Voting of students matters

Tübingen, Germany

10/2018 - 04/2021

FSK Tübingen

MEMBER OF THE STUDENT COUNCIL OF COGNITIVE SCIENCE

- Representing student interests
- Communication between students and instructors
- Organizing social events

Tübingen, Germany

10/2018 - 04/2021

Interests

I am passionate about learning new languages and exploring different cultures. To disconnect, I enjoy hiking in nature and spending quality time with family and loved ones.